Unit 07: Data Preprocessing

CONTENTS

Objectives

After this lecture, you will be able to

- Learn the need for preprocessing of data.
- Know the major Tasks in Data Preprocessing
- Understand the concept of missing data and various methods for handling missing data.
- Learn the concept of data discretization and data reduction.
- Understand the various strategies of data reduction.

Introduction

Preprocessing data is a data mining technique for transforming raw data into a usable and efficient format. The measures taken to make data more suitable for data mining are referred to as data preprocessing. The steps involved in Data Preprocessing are normally divided into two categories:

- Selecting data objects and attributes for analysis, and selecting data objects and attributes for analysis.
- Adding/removing attributes is a two-step process.

In data mining, data preprocessing is one of the most important aspects of the well-known information innovation from the data processor. Data taken directly from the source will contain errors, contradictions, and, most importantly, will not be willing to be used for a data mining tool.

7.1 Why Data Preprocessing?

We need to preprocess the data because of the following reasons:

1. Data in the real world is dirty which means the data is :
    - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - noisy: containing errors or outliers
    - inconsistent: containing discrepancies in codes or names

2. No quality data, no quality mining results!
    - Quality decisions must be based on quality data
    - Data warehouse needs consistent integration of quality data

Multi-Dimensional Measure of Data Quality

A well-accepted multidimensional view must have the following qualities:

- Accuracy
- Completeness
- Consistency
- Timeliness
- Believability
- Value-added
- Interpretability
- Accessibility

Accuracy: The degree to which knowledge accurately represents an event or object mentioned is referred to as "accuracy."

Example, If a customer's age is 32 but the system says she's 34, the system is inaccurate.

Completeness: When data meets comprehensiveness expectations, it is considered "complete." Assume you ask the customer to include his or her name. You can make a customer's middle name optional, but the data is full as long as you have their first and last names.

Consistency: The same information can be held in many locations at several businesses. It's called "consistent" if the information matches.

Example: ,It's inconsistent if your human resources information systems claim an employee no longer works there, but your payroll system claims he's still getting paid.

Timeliness: Is your data readily accessible when you need it? The "timeliness" factor of data quality is what it is called. Let's say you need financial data every quarter; the data is timely if it arrives when it's expected to.

7.2 Major Tasks in Data Preprocessing

The following diagram shows the various task associated with data preprocessing:

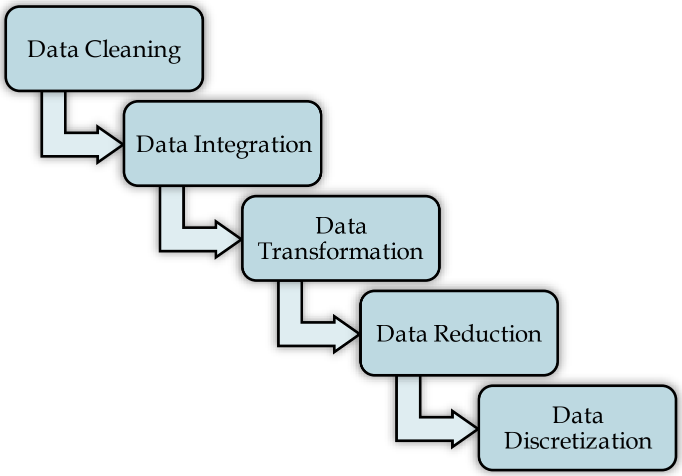

Figure 1:Data Preprocessing Tasks

Data Cleaning: Data cleaning is the method of cleaning data to make it easier to integrate.

Data Integration: Data integration is the method of bringing all of the data together.

Transformation of data: The process of transforming data into a reliable format is known as data transformation.

Reduction of data: Data reduction is the method of breaking down large amounts of data into smaller chunks so that they can be easily converted.

Discretization of data: Data discretization reduces a large number of data values to a limited number of them, making data assessment and management much easier.

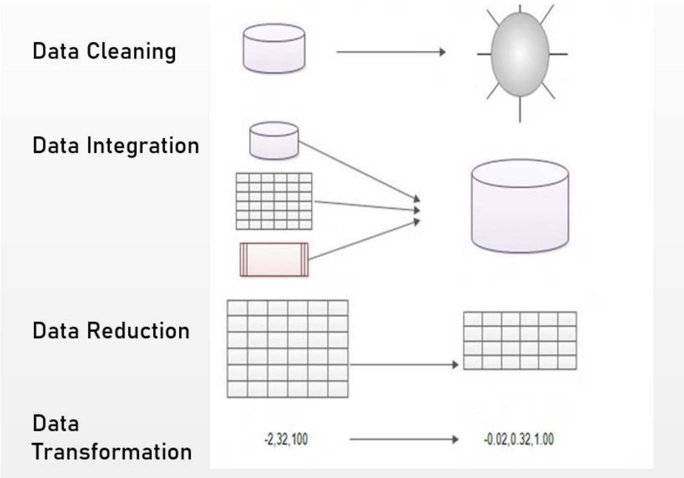

Figure 2: Forms of Data Preprocessing

Data Cleaning Task

When it comes to the final review, the quality of the data is crucial. Any data that is incomplete, noisy, or inconsistent may have an impact on your final result. The process of detecting and deleting corrupt or incorrect records from a record collection, table, or database is known as data cleaning in data mining. The following are the data cleaning tasks:

- Fill in missing values
- Identify outliers and smooth out noisy data
- Correct inconsistent data

Missing data is not always available.Missing data may be due to

- equipment malfunction
- inconsistent with other recorded data and thus deleted
- data not entered due to misunderstanding
- certain data may not be considered important at the time of entry

Example: many tuples have no recorded value for several attributes, such as customer income in sales data

Some data cleansing techniques:

1 Forget the tuple. If the classmark is absent, this is finished. Unless the tuple includes multiple attributes with missing values, this approach is ineffective.

2 You can manually fill in the missing value. This method works well with small data sets that have some missing values.

3. You may use a global constant like "Unknown" or minus infinity to replace all missing attribute values.

4 To fill in the missing value, use the attribute mean. If a customer's average income is $25,000, you can use this amount to fill in the lost income value.

5 Fill in the missing value with the most likely value.

Noisy Data

A random error or deviation in a calculated variable is referred to as noise. Noisy data may occur as a result of defective data collection instruments, data entry issues, or technological limitations.

How Do You Deal With Noisy Data?

Binning: Binning approaches sorted data values by consulting their "neighborhood," or the values in their immediate vicinity. The sorted values are divided into several "buckets" or bins.

Equal-depth (frequency) partitioning:

- It divides the range into N intervals, each containing the approximately same number of samples
- Good data scaling
- Managing categorical attributes can be tricky.

Equal-width (distance) partitioning:

- It divides the range into N intervals of equal size: uniform grid
- if A and B are the lowest and highest values of the attribute, the width of intervals will be $W = \frac{B-A}{N}$.
- The most straightforward
- But outliers may dominate the presentation
- Skewed data is not handled well.

As an example,

Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

- Partition into (equi-depth) bins:

- Bin 1: 4, 8, 9, 15

- Bin 2: 21, 21, 24, 25

- Bin 3: 26, 28, 29, 34

- Smoothing by bin means:

- Bin 1: 9, 9, 9, 9

- Bin 2: 23, 23, 23, 23

- Bin 3: 29, 29, 29, 29

- Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15

- Bin 2: 21, 21, 25, 25

- Bin 3: 26, 26, 26, 34

Data Integration Task

Data integration is a data preprocessing technique that involves integrating data from several heterogeneous data sources into a single data store. Multiple data cubes, databases, and flat files are examples of these sources.

Data mining combines techniques from a variety of fields, including database and data warehouse technology, statistics, machine learning, high-performance computing, pattern recognition, neural networks, data visualisation, information retrieval, image and signal processing, and spatial and temporal data analysis

For data integration, there are primarily two approaches:

A. **Tight Coupling**

The method of ETL - Extraction, Transformation, and Loading - is used to integrate data from various sources into a single physical location in tight coupling.

B. **Loose Coupling**

Loose Coupling is a term that refers to a coupling that is only the data from the source databases is held in loose coupling. An interface is given in this method that takes a user's query and converts it into a format that the source database can understand, then sends the query directly to the source databases to get the response.

Data Integration Issues

We must deal with many issues while integrating the data, which are discussed below:

1. **Entity Identification Problem**

How do we 'match the real-world entities from the data, given that the data is unified from heterogeneous sources? We have consumer data from two separate data sources, for example. The customer id is assigned to one data source's entity, while the customer number is assigned to the other data source's entity. How does the data analyst or the machine know that these two entities lead to the same thing?

2. **Redundancy and Correlation Analysis**

One of the major concerns during data integration is redundancy. Unimportant data or data that is no longer needed is referred to as redundant data. It can also happen if there are attributes in the data set that can be extracted from another attribute.

Example: If one data set contains the customer's age and another data set contains the customer's date of birth, age will be a redundant attribute since the date of birth could be used to derive it.

The extent of redundancy is also increased by attribute inconsistencies. Correlation analysis can be used to find the redundancy. The attributes are evaluated to see whether they are interdependent on one another and if so, to see if there is a connection between them.

3. **Data Conflict Detection and Resolution**

Data conflict occurs when data from various sources is combined and does not fit. The attribute values, for example, can vary between data sets. The disparity maybe because they are depicted differently in different data sets. Assume that the price of a hotel room in different cities is expressed in different currencies. This type of problem is detected and fixed during the data integration process.

Data Transformation Task

Data is converted from one format to another which is more suitable for data mining during the data transformation process.

Listed below are a few data transformation strategies:-

1. **Smoothing:** The term "smoothing" refers to the process of removing noise from data.
2. **Aggregation:**Aggregation is the application of description or aggregation operations to data.
3. **Generalization:** Using definition hierarchies climbing, low-level data is replaced with high-level data in generalization.
4. **Normalization:** Using normalization, attribute data was scaled to fall within a narrow range, such as 0.0 to 1.0.
5. **Attribute Construction:** New attributes are created from a specified collection of attributes in attribute construction.

Data Smoothing

Smoothing data entails eliminating noise from the data collection under consideration. We've seen how techniques like binning, regression, and clustering are used to eliminate noise from results.

Binning: This approach divides the sorted data into several bins and smooths the data values in each bin based on the values in the surrounding neighborhood.

Regression: This approach determines the relationship between two dependent attributes so that we can use one attribute to predict the other.

Clustering: In this process, related data values are grouped together to create a cluster. Outliers are values that are located outside of a cluster.

Aggregation of Data

By performing an aggregation process on a large collection of data, data aggregation reduces the volume of the data set.

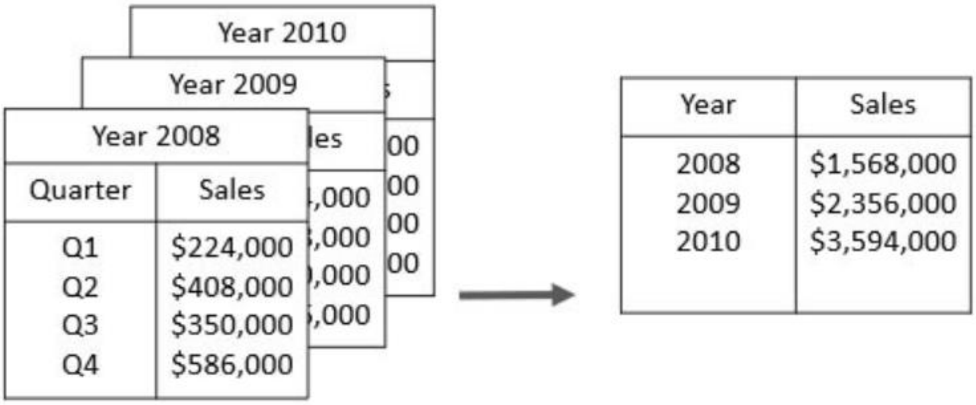

Figure 3: Aggregated Data

Example: we have a data set of sales reports of an enterprise that has quarterly sales of each year. We can aggregate the data to get the annual sales report of the enterprise.

Generalization

The nominal data or nominal attribute is a set of values with a finite number of unique values and no ordering between them. Job-category, age-category, geographicregions, items-category, and other nominal attributes are examples. By adding a group of attributes, the nominal attributes form the definition hierarchy. Concept hierarchy can be created by combining words like street, city, state, and nation.

The data is divided into several levels using a concept hierarchy. At the schema level, the definition hierarchy can be created by adding partial or complete ordering between the attributes. Alternatively, a concept hierarchy can be created by specifically grouping data on a portion of the intermediate level.

Normalization

The process of data normalization entails translating all data variables into a specific set. Data normalization is the process of reducing the number of data values to a smaller range, such as [-1, 1] or [0.0, 1.0].

The following are some examples of normalization techniques:

A. Min-Max Normalization:

Minimum-Maximum Normalization is a linear transformation of the original results. Assume that the minima and maxima of an attribute are min A and max A, respectively.

We Have the Formula:

$$v' = \frac{v - min_P}{max_P - min_P}(new\_max_P - new\_min_P) + new\_min_P$$

Where v is the value in the new range that you want to plot. After normalizing the old value, you get v', which is the new value. For example, the minimum and maximum values for the attribute 'income' are \$1200 and \$9800, respectively, and the range in which we would map a value of \$73,600 is [0.0, 1.0]. The value of \$73,600 will be normalized using the min-max method:

$$\frac{73600-1200}{9800-1200}(1.0-0.0)+0.0=0.716$$

B. Z-score Normalization

Using the mean and standard deviation, this approach normalizes the value for attribute A. The formula is as follows:

$$v'_i=\frac{v_i-\bar{A}}{\sigma_A}$$

Here $\bar{A}$ and $\sigma A$are the mean and standard deviation for attribute A are respectively.For instance, attribute A now has a mean and standard deviation of \$54,000 and \$16,000, respectively. We must also use z-score normalization to normalize the value of \$73,600.

$$\frac{73600-5400}{1600}=1.225$$

C. Decimal Scaling

By shifting the decimal point in the value, this method normalizes the value of attribute A. The maximum absolute value of A determines the movement of a decimal point.

The decimal scaling formula is as follows:

$$v'_i=\frac{v_i}{10^j}$$

Here j is the smallest integer such that max(|v'i|)<1.

For example, the observed values for attribute A range from -986 to 917, with 986 as the maximum absolute value. To use decimal scaling to normalize each value of attribute A, we must divide each value of attribute A by 1000, i.e. j=3. As a result, the value -986 is normalized to -0.986, and the value 917 is normalized to 0.917.

To uniformly normalize future data, normalization parameters such as mean, standard deviation, and maximum absolute value must be maintained.

Attribute Construction

In the attribute construction process, new attributes are created by consulting an existing collection of attributes, resulting in a new data set that is easier to mine. Consider the following scenario: we have a data set containing measurements of various plots, such as the height and width of each plot. So, using the attributes 'height' and 'weight,' we can create a new attribute called 'area.' This often aids in the comprehension of the relationships between the attributes in a data set.

7.3 Data Discretization

Data discretization encourages data transformation by replacing integer data values with interval marks. The interval labels (0-10, 11-20...) or the interval labels (0-10, 11-20...) may be used to substitute the values for the attribute "age" (kid, youth, adult, senior).Data discretization can be classified into two types **supervised discretization** where the class information is used and the other is **unsupervised discretization** which is based on which direction does the process proceed i.e. 'top-down splitting strategy' or 'bottom-up merging strategy'. For Discretization different attributes needs consideration which is as follows:

Nominal Attribute: Nominal Attributes only provide enough information to distinguish one object from another. For example, the person's sex and the student's roll number.

Ordinal Attribute: The value of the ordinal attribute is sufficient to order the objects. Rankings, grades, and height are only a few examples.

Numeric Attributes: It is quantitative, in the sense that quantity can be calculated and expressed in integer or real numbers.

Ratio Scaled attribute: A proportion Ratio is a scaled parameter that is important for both inequalities and ratios. For example, age, height, and weight.

Types of Data Discretization

Top-down Discretization

Top-down discretization or splitting is when a process begins by finding one or a few points (called breakpoints or cut points) to split the entire attribute set, and then repeats the process recursively on the resulting intervals.

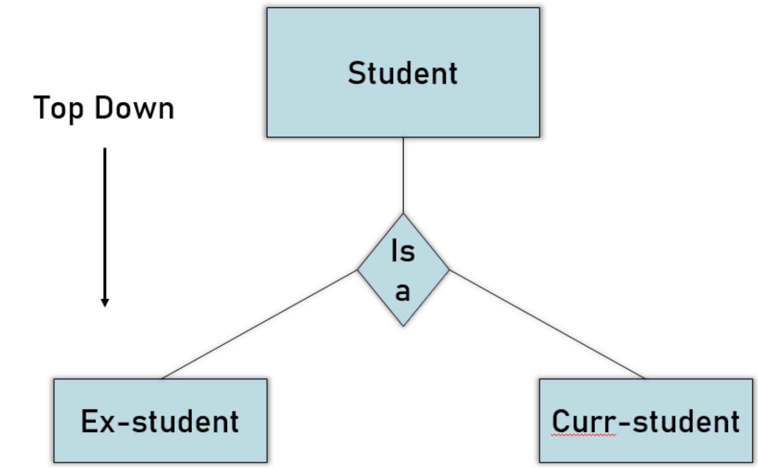

Figure 4: Top-Down Discretization

Bottom-up Discretization

Bottom-up discretization or merging is described as a process that begins by considering all continuous values as potential split-points and then removes some by merging neighborhood values to form intervals.

Discretization on an attribute can be done quickly to create a definition hierarchy, which is a hierarchical partitioning of the attribute values.

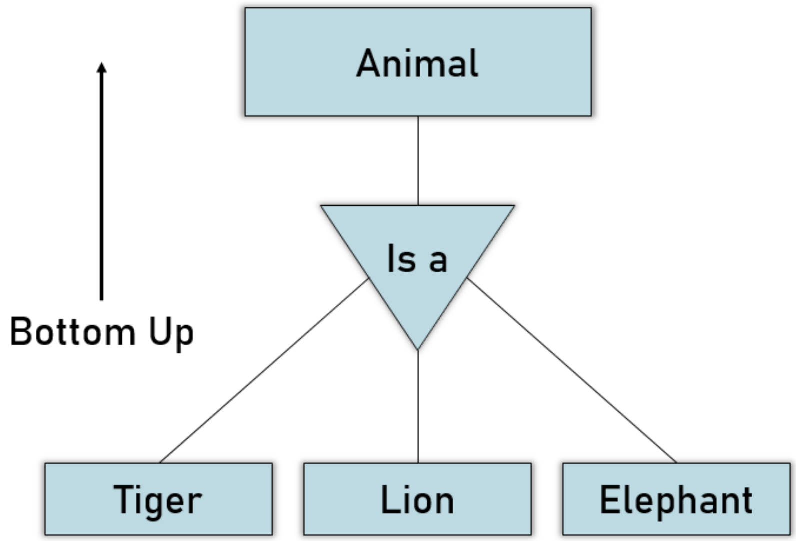

Figure 5: Bottom-Up Discretization

Supervised Discretization

Supervised discretization is when you take the class into account when making discretization boundaries.The discretization must be determined solely by the training set and not the test set.

Unsupervised Discretization

Unsupervised discretization algorithms are the simplest algorithms to make use of, because the only parameter you would specify is the number of intervals to use; or else, how many values should be included in each interval.

Discuss the significance of supervised and un-supervised discretization.

7.4 Need of Data Reduction

Terabytes of data may be stored in a database or data center. As a result, data processing and mining on such massive datasets can take a long time.

Data reduction methods may be used to create a smaller-volume representation of the data set that still contains important information. The data reduction process reduces the size of data and makes it suitable and feasible for analysis. In the reduction process, the integrity of the data must be preserved, and data volume is reduced. Many strategies can be used for data reduction.

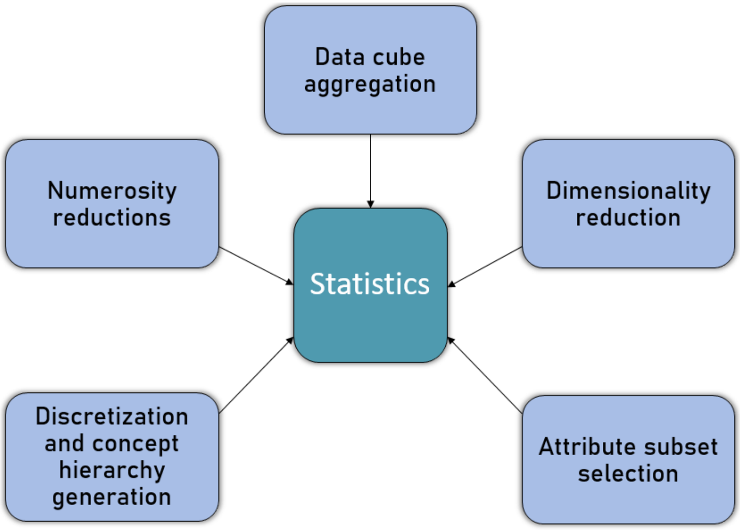

Figure 6: Data Reduction Strategies

1. **Data Cube Aggregation:** In the construction of a data cube, aggregation operations are applied to the data. This method is used to condense data into a more manageable format. As an example, consider the data you collected for your study from 2012 to 2014, which includes your company's revenue every three months. Rather than the quarterly average, they include you in the annual revenue.

2. **Dimension Reduction:**We use the attribute needed for our analysis whenever we come across data that is weakly significant. It shrinks data by removing obsolete or redundant functions. The following are the methods used for Dimension reduction:

- Step-wise Forward Selection
- Step-wise Backward Selection
- Combination of forwarding and Backward Selection

Step-wise Forward Selection

The selection begins with an empty set of attributes later on we decide the best of the original attributes on the set based on their relevance to other attributes.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: { }

- Step-1: {X1}

- Step-2: {X1, X2}
- Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

Step-wise Backward Selection

This selection starts with a set of complete attributes in the original data and at each point, it eliminates the worst remaining attribute in the set.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: {X1, X2, X3, X4, X5, X6 }

- Step-1: {X1, X2, X3, X4, X5}
- Step-2: {X1, X2, X3, X5}
- Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

Combination of forwarding and backward selection

It helps us to eliminate the worst attributes and pick the better ones, saving time and speeding up the process.

Data Compression

Using various encoding methods, the data compression technique reduces the size of files (Huffman Encoding & run-length Encoding). Based on the compression techniques used, we can split it into two forms.

- Lossless Compression
- Lossy Compression

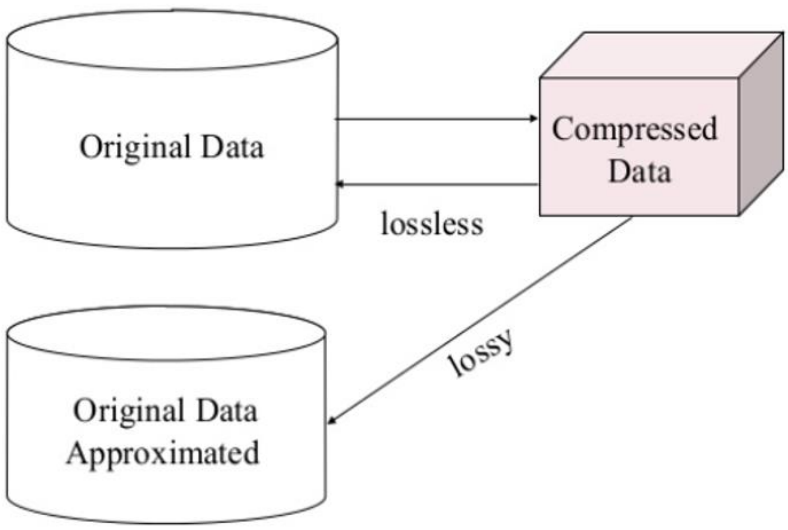

Figure 7: Data Compression Techniques

Lossless Compression – Encoding techniques (Run Length Encoding) allow for a quick and painless data reduction. Algorithms are used in lossless data compression to recover the exact original data from the compressed data.

Lossy Compression – Examples of this compression include the Discrete Wavelet Transform Technique and PCA (principal component analysis). JPEG image format, for example, is a lossy compression, but we can find a sense equivalent to the original image. The decompressed data in lossy data compression vary from the original data, but they are still useful for retrieving information.

Numerosity Reduction

Numerosity Reduction is a data reduction strategy that uses a smaller type of data representation to replace the original data. There are two approaches for reducing numerosity: parametric and non-parametric.

Parametric Methods: Parametric methods use a model to represent data. The model is used to estimate data, requiring only data parameters to be processed rather than real data. These models are built using regression and log-linear methods.

- **Regression**

Easy linear regression and multiple linear regression are two types of regression. While there is only one independent attribute, the regression model is referred to as simple linear regression, and when there are multiple independent attributes, it is referred to as multiple linear regression.

- **Log-Linear Model**

  Based on a smaller subset of dimensional combinations, a log-linear model may be used to estimate the likelihood of each data point in a multidimensional space for a collection of discretized attributes. This enables the development of a higher-dimensional data space from lower-dimensional attributes.

Did you Know? On sparse data, both regression and the log-linear model can be used, but their implementation is limited.

Non-Parametric Methods: Histograms, clustering, sampling, and data cube aggregation are examples of non-parametric methods for storing reduced representations of data.

- **Histograms:** A histogram is a frequency representation of data. It is a common method of data reduction that employs binning to approximate data distribution.
- **Clustering:** Clustering is the division of data into classes or clusters. This method divides all of the data into distinct clusters. The cluster representation of the data is used to replace the actual data in data reduction. It also aids in the detection of data outliers.
- **Sampling:** Sampling is a data reduction technique that allows a large data set to be represented by a much smaller random data set.
- **Aggregation of Data Cubes**: Aggregation of data cubes entails transferring data from a complex level to a smaller number of dimensions. The resulting data set is smaller in size while retaining all of the details required for the analysis mission.

Summary

- Data transformation is the process of converting data into a format that allows for effective data mining.
- Normalization, discretization, and concept hierarchy are the most powerful ways of transforming data.
- Data normalization is the process of growing the collection of data.
- The data values of a numeric variable are replaced with interval labels when data discretization is used.
- The data is transformed into multiple l by using concept hierarchy.
- Accuracy, completeness, continuity, timeliness, believability, and interpretability are all w ords used to describe data quality. These qualities are evaluated based on the data's intend ed use.
- Data cleaning routines aim to fix errors in the data by filling in missing values, smoothing out noise when finding outliers, and filling in missing values. Cleaning data is normally done in two steps, one after the other.

Keywords

Data cleaning: To remove noise and inconsistent data.

Data integration: Multiple data sources may be combined.

Data transformation: Where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.

Data discretization: Ittransforms numeric data by mapping values to interval or concept

labels.

Self Assessment Questions

1. What is the use of data cleaning?

A. To remove the noisy data

B. Correct the inconsistencies in data

C. Transformations to correct the wrong data.

D. All of the above

2. Data cleaning is

A. Large collection of data mostly stored in a computer system

B. The removal of noise errors and incorrect input from a database

C. The systematic description of the syntactic structure of a specific database. It describes the structure of the attributes of the tables and foreign key relationships.

D. None of these

3. Which of the following is not a data pre-processing methods

A. Data Visualization

B. Data Discretization

C. Data Cleaning

D. Data Reduction

4. Dimensionality reduction reduces the data set size by removing _________

A. composite attributes

B. derived attributes

C. relevant attributes

D. irrelevant attributes

5. Which of the following activities is a data mining task?

A. Monitoring the heart rate of a patient for abnormalities

B. Extracting the frequencies of a sound wave

C. Predicting the outcomes of tossing a (fair) pair of dice

D. Dividing the customers of a company according to their profitability

6. Which of the following is NOT an example of ordinal attributes?

A. Ordered numbers

B. Movie ratings

C. Zipcodes

D. Military ranks

7. Which data mining task can be used for predicting wind velocities as a function of temperature, humidity, air pressure, etc.?

A. Cluster Analysis

B. Regression

C. Classification

D. Sequential pattern discovery

8. Which of the following is not a data mining task? Select one:

A. Feature Subset Detection

B. Association Rule Discovery

C. Regression

D. Sequential Pattern Discovery

9. ______________Combines data from multiple sources into a coherent store.

A. Data integration

B. Data reduction

C. Data Transformation

D. Data cleaning

10. Careful integration can help reduce and avoid __________and inconsistencies in resulting data set.

A. Noise

B. Redundancies

C. Error

D. None

11. _________________is the process of changing the format, structure, or values of data.

A. Data integration

B. Data reduction

C. Data Transformation

D. Data cleaning

12. In Binning, we first sort data and partition into (equal-frequency) bins and then which of the following is not a valid step

A. smooth by bin boundaries

B. smooth by bin median

C. smooth by bin means

D. smooth by bin values

13. Which of the following is NOT a data quality-related issue?

A. Missing values

B. Outlier records

C. Duplicate records

D. Attribute value range

14. Which of the following is used to remove noise from data.

A. Generalization

B. Normalization

C. Aggregation

D. Smoothing

15. IN ________________ Encoding mechanisms are used to reduce the data set size.

A. Numerosity Reduction

B. Data Compression

C. Dimensionality Reduction

D. Data Cube Aggregation

Answers for Self Assessment

1.	D	2.	B	3.	A	4.	D	5.	A
6.	C	7.	B	8.	A	9.	A	10.	B
11.	C	12.	D	13.	C	14.	D	15.	B

Review Questions

Q1) Data quality can be assessed in terms of several issues, including accuracy, completeness, and consistency. For each of the above three issues, discuss how data quality assessment can depend on the intended use of the data, giving examples. Propose two other dimensions of data quality.

Q2) In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem.

Q3) Discuss issues to consider during data integration.

Q4) For example explain the various methods of Normalization.

Q5) Elaborate on various data reduction strategies by giving the example of each strategy.

Q6) Explain the concept of data discretization along with its various methods.

Q7) What is the need for data preprocessing? Explain the different data preprocessing tasks in detail.

Further Readings

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